

Stochastic Model Capstone Design

- Final Presentation



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Project Review

Project

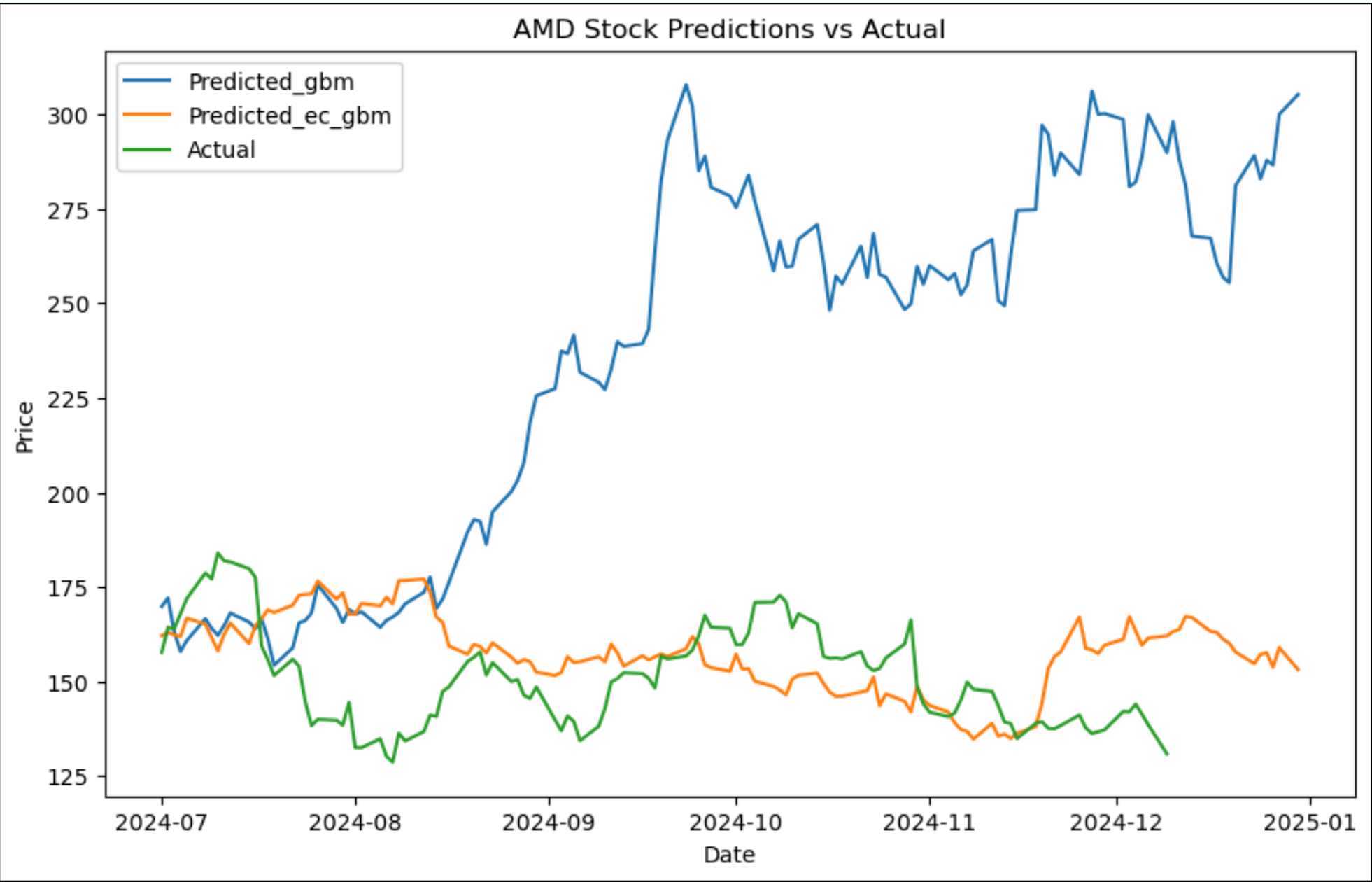
Hyperparameter Tunning

Modified EC–GBM Model

Model Evaluation

Project Review

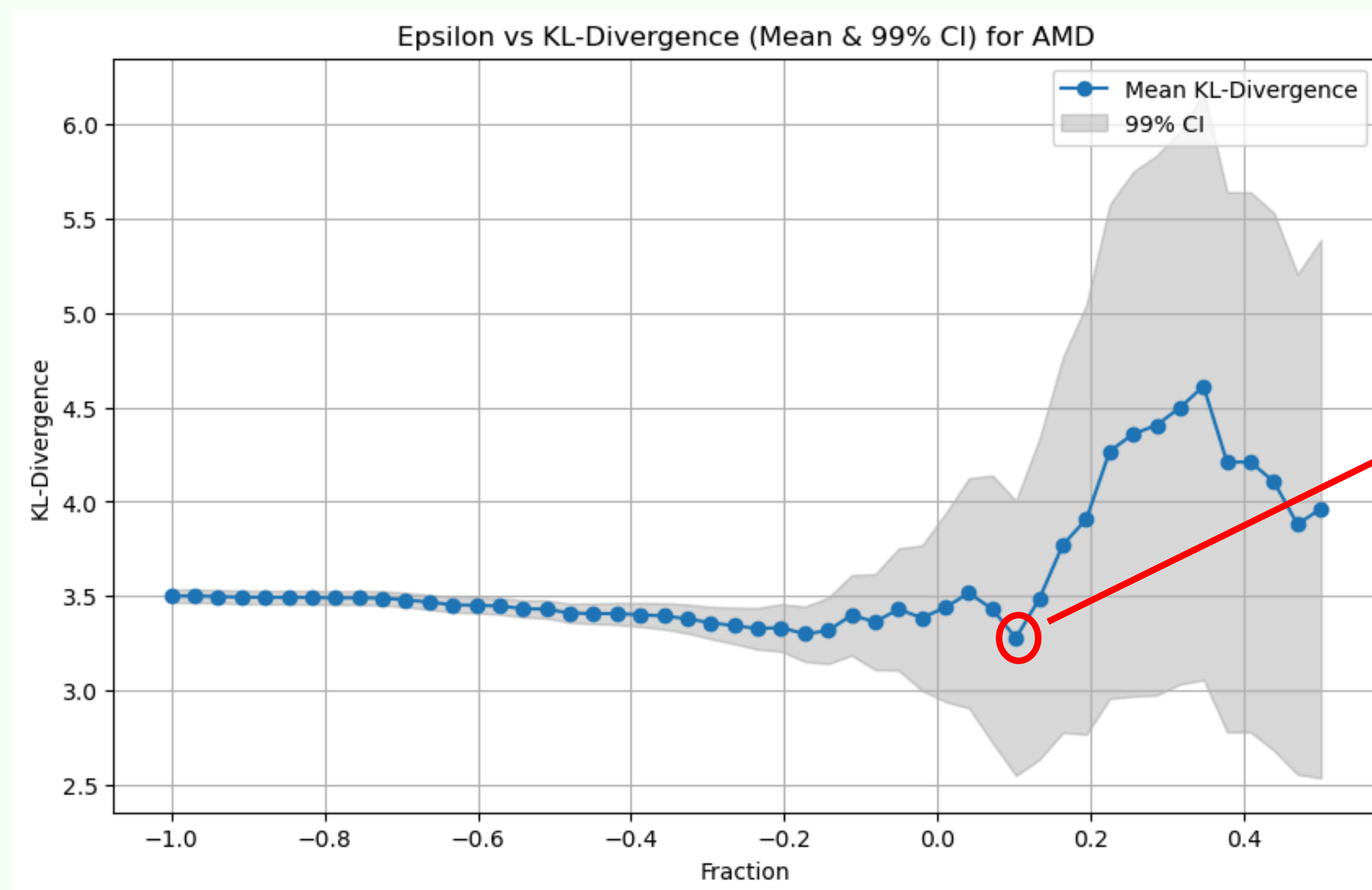
Last week, we conducted stock price predictions using GBM and EC-GBM and evaluated the performance of the two models.



AMD 	MAE	MSE	D _{KL}
GBM	82.28	9111.14	9.61
EC-GBM	14.61	338.67	6.98

Hyperparameter Tunning

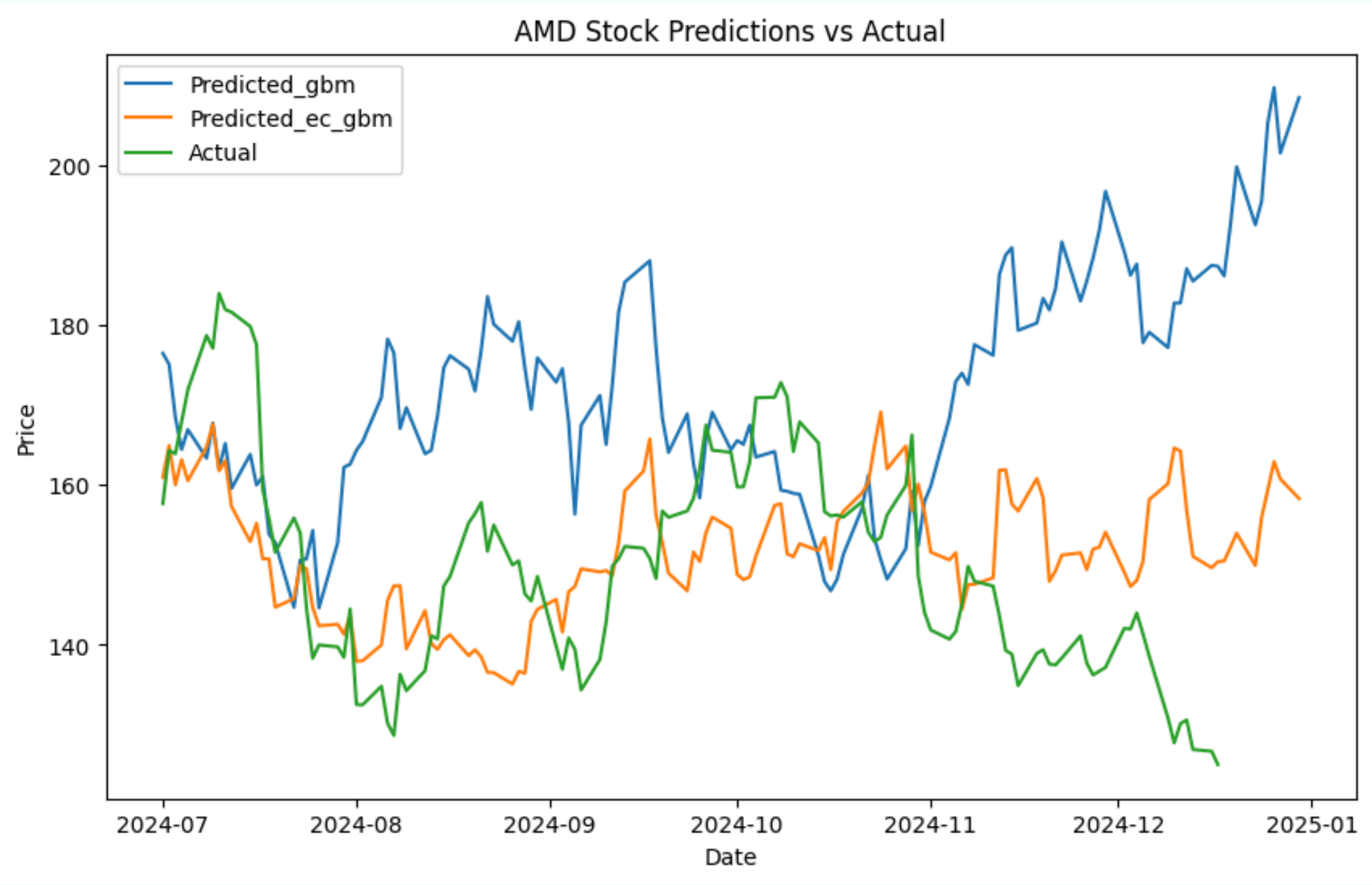
To further enhance the performance of the existing EC-GBM model, we will determine the **epsilon** value using KLD.



FRACTION : 0.102041
EPSILON : 0.004056

Modified EC-GBM Model

Using the optimal epsilon determined, we modified EC-GBM model to improve its performance.



AMD 	MAE	MSE	D _{KL}
GBM	24.22	868.58	7.74
EC-GBM	11.14	181.93	4.24

Model Evaluation

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EC-GBM	11.14	181.93	4.24

USING THE OPTIMAL EPSILON AS A HYPERPARAMETER,
WE OBSERVED IMPROVEMENTS IN ALL THREE EVALUATION METRICS.

Final

WHILE CARRYING OUT THE EC-GBM PROCESS, GENERATING RANDOM WALKS TO FIND ENTROPY-BASED OPTIMAL PATHS TOOK EXCESSIVELY LONG. THEREFORE, A NEW APPROACH IS NEEDED TO DRASTICALLY REDUCE THE RANDOM WALK GENERATION TIME. ADDITIONALLY, RUNNING THE MODEL IN A COLAB ENVIRONMENT UNDERSCORED THE LIMITATIONS OF AVAILABLE COMPUTING RESOURCES.

FOR FUTURE RESEARCH, BY MAXIMIZING THE COMPUTATIONAL EFFICIENCY OF THE ENTROPY-BASED PATH EXPLORATION PROCESS AND ACTIVELY UTILIZING HIGH-PERFORMANCE COMPUTING RESOURCES AND APPROXIMATION ALGORITHMS, IT IS EXPECTED THAT THE EC-GBM MODEL CAN DELIVER FASTER AND MORE RELIABLE PREDICTIVE PERFORMANCE ACROSS A WIDE RANGE OF ENVIRONMENTS.

Reference

GUPTA, R., DRZAZGA-SZCZĘŚNIAK, E.A., KAIS, S. ET AL. ENTROPY CORRECTED GEOMETRIC BROWNIAN MOTION. SCI
REP 14, 28384 (2024). [HTTPS://DOI.ORG/10.1038/S41598-024-79714-3](https://doi.org/10.1038/s41598-024-79714-3)

THANK YOU